Stochastic Averaging Method for Periodically driven systems: Application to cross-frequency coupling.

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Abstract

Cross-frequency Phase-Amplitude coupling (PAC) is a widespread phenomenon observed in neural rhythm recordings across species and brain areas. It describes the modulation of the amplitude of a fast rhythm by the phase of a slow one. It has received considerable attention for its potential roles in cognitive processes, including learning, working memory, and communication between distant brain regions. However, few modeling and theoretical studies allow us to understand how fast and slow rhythms interact to produce Phase-Amplitude coupling. Here, we provide such a theory. Specifically, we begin with an Excitatory-Inhibitory network oscillating in the gamma frequency band (30-100 Hz), subject to slow exogenous inputs, and decompose its envelope and phase. In contrast to previous modeling approaches, our approach incorporates the dynamical regimes (noise-induced and noise-perturbed rhythms) of gamma rhythms simultaneously. It integrates biological parameters of the E-I neural network. Our theory suggests that PAC depends on the dynamical regime of the E-I network and is achieved by increasing only the amplitude of the external inputs. The slow external signal modulates the fundamental component of the eigenvalue of the fast rhythms, thereby creating amplitude modulation. Moreover, our theory allows multiplexing of PAC, in which several slow external inputs induce PAC. Finally, the dynamics of the fast rhythms evolve on top of a slow-frequency time-varying baseline. The theory can be extended to the most realistic case where other brain areas generate the slow inputs. The results obtained represent a step towards the understanding of the mechanisms and functions of CFC in terms of networks and exogenous input parameters for biological and non-biological rhythms.

Introduction

Neural rhythms are ubiquitously observed in different brain areas of other species [6]. They are usually the mixing of two or more fundamental rhythms, each oscillating within a given frequency band, a mechanism known as cross-frequency coupling (CFC). This is the case for rhythms in the rodent hippocampus and the human neocortex [9], which show coupling between theta (3-8 Hz) and gamma (30-100 Hz) bands. CFC is also seen in other species, including anesthetized and awake rats [4, 26, 27], and the auditory cortex of monkeys during spontaneous and stimulus-driven activity [16]. Moreover, CFC has been shown to depend on tasks such as long-term memory and working memory [20]. This suggests that CFC may have a functional role in the brain. Investigations into the role of CFCs are ongoing research topics. Previous studies argued that the brain uses CFC for multi-item working memory [2, 15, 17], speech encoding [13], or communication between distant areas [7]. The potential roles of CFC are linked to their different signatures. Since oscillations can be decomposed into amplitude, phase, and frequency components, their coupling across frequencies can also be interpreted as the coupling among these components. Then, distinct signatures of CFC refer to whether the rhythms are coupled through their amplitudes (Amplitude-Amplitude coupling or AAC), phases (Phase-Phase coupling or PPC), the frequency of the fast with the phase of the slow (Phase-Frequency coupling or PFC), and the amplitude of the fast with the phase of the slow rhythms (Phase-Amplitude coupling or PAC). Other combinations are not observed in the data. Several techniques have been developed to quantify the distinct CFC signatures [11, 29]. However, few theoretical studies have addressed their modeling. In other words, how should fast and slow rhythms be mixed to produce different signatures of CFC? What

is the effect of the dynamical regime of the fast rhythm on the CFC? In this work, we focus on the PAC because it is prevalent in the data.

We consider a stochastic Wilson-Cowan (SWC) neural network of a Pyramidal-Interneuron (E-I) Network Gamma (PING) type, subject to slow exogenous inputs. We consider cases in which the SWC generates noise-induced or noise-perturbed fast rhythms before interacting with the slow external inputs. Numerical simulations of the SWC Local Field potentials (LFPs) power spectra show higher and lower peaks, suggesting coupling between fast and slow rhythms. Our goal is to theoretically investigate the types of coupling involved in the dynamics of E-I LFPs and to study how different network and external input parameters shape the strength and behaviour of the PAC. Technically, we perform a Taylor expansion of the SWC to lowest order and identify two principal types of interaction between the SWC LFPs and the slow exogenous inputs: parametric and additive couplings. To understand the role of each type of coupling, we assumed that parametric couplings contribute only to the fast component of the SWC LFPs (i.e., the higher peaks in the power spectra). In contrast, additive couplings describe the slow component (i.e., the lower peaks in the power spectra). We seek reduced expression of the fast and slow components (or fast and slow rhythms) separately. For the fast rhythms described by parametric couplings alone, we perform an envelope-phase decomposition using the Stochastic Averaging Method (SAM). The results show that slow external inputs modulate both the envelope and the phase dynamics of the fast rhythms, thereby enabling PAC. Therefore, parametric couplings are responsible for PAC. Using our envelope-phase decomposition, we studied PAC as a function of different parameters, including the modulation index (MI) [29]. In fact, the PAC behaviour depends on the dynamical regime (noise-induced or noise-perturbed) of the former SWC. It is readily observed as the amplitudes of the exogenous inputs increase. Multiplexing CFC, where several slow external inputs interact with the fast rhythms, is also described through our reduced envelope-phase equations. The reduced dynamics of the slow rhythms (the slow component of the SWC LFPs) obtained from additive couplings alone are computed using linear algebra techniques. The slow rhythms can be interpreted as a time-varying baseline. Then, the dynamics of the SWC under exogenous inputs can be approximated as the sum of the dynamics induced by both parametric and additive couplings. Our reduced solutions are in good agreement with the former SWC models in the noise-induced regime and for weak exogenous input amplitudes.

Previous work has shown that harmonic waves can describe the outputs of oscillatory networks [23], as in the external inputs described here. Our theory can then be readily extended to more realistic cases in which other brain areas generate external inputs. Similarly, we can use our framework to describe CFC data in vivo. Specifically, SWC LFPs may be matched to raw data, with their fast and slow components corresponding to passband-filtered components of the raw data in the rapid (gamma) and slow (theta) frequency bands, respectively. The results obtained here provide a sound theoretical basis for understanding the Phase-Amplitude coupling mechanisms and the roles they may play in brain function and other systems in which PAC is observed.

Results

1 The Model of Intrinsic Gamma Rhythms

Phase-Amplitude coupling describes the mechanism by which the phase of a slower rhythm modulates the amplitude of a faster one. In the context of neural rhythms, the faster rhythm is usually assumed to be locally generated by a neural network. To mimic the fast rhythm, here we consider the stochastic Wilson Cowan (SWC) model [32] described as follows: the mean synaptic coupling from E cells to themselves, and from E cells to I cells, are given respectively by W_{EE} and W_{IE}. Similarly, the mean synaptic coupling from I cells to themselves and from I cells to E cells are W_{II} and W_{EI}. The E cell population receives a constant external stimulus h_E, while the I cell population is driven by the continuous external current h_I (Fig.1-**(A)**). The two $E-I$ populations are subjected to their random environment modelled as two independent Gaussian white noises $\xi_E(t)$ and $\xi_I(t)$ with intensities σ_E and σ_I

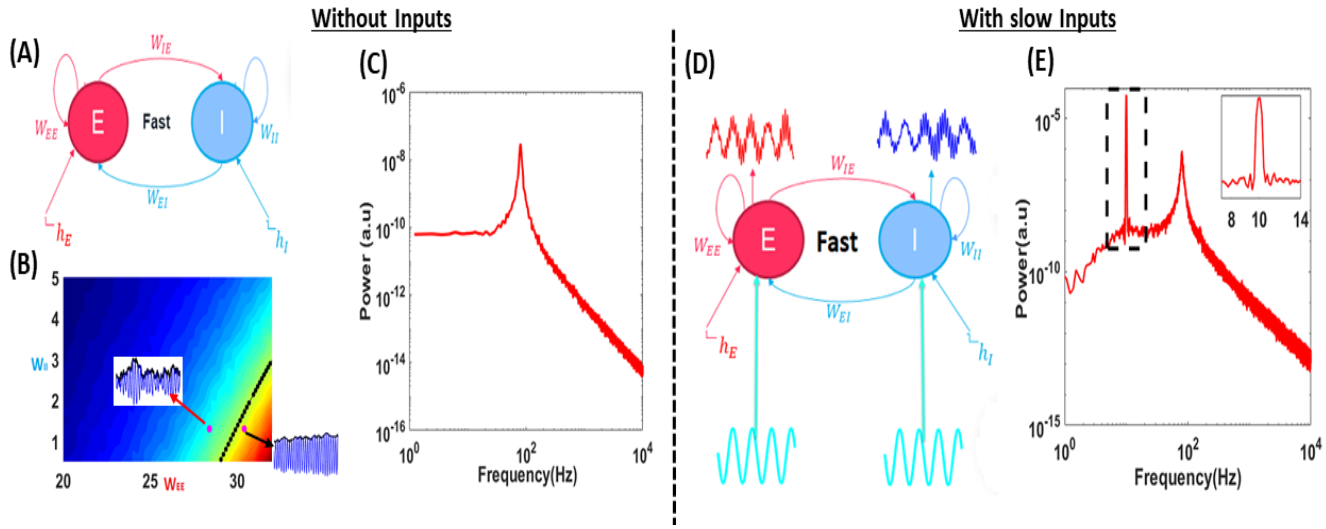

Figure 1. Graphical illustration of a PING and its corresponding power spectra with and without the slow external inputs. **(A)** Cartoon of the PING with only constant external inputs.**(B)**: Bifurcation diagram of the network described in **(A)** in the recurrent (W_{EE}, W_{II}) subspace. Two distinct regimes separated by a black dotted line are shown. From left to right, before the bifurcation line, oscillations are noise-induced (see inset). After the bifurcation, rhythms are noise-perturbed. **(C)** Power spectrum of the E-LFP shows a single peak around 85 Hz reflecting the intrinsic frequency of the network. **(D)**: Cartoon of the PING subject to slow (10 Hz) external inputs. **(E)**: The power spectrum of the E-LFP shows two peaks, one around 85 Hz reflecting the contribution of the intrinsic PING and another one at 10 Hz (see the inset) showing the interaction with the external inputs. In the two examples involving the power spectra **(A-B)**, the fast gamma rhythms fall within the noise-induced regime described in **(B)**. .

respectively. A mathematical formulation can be given as:

$$\frac{dE(t)}{dt} = -\alpha_E E(t) + (1 - E(t))\beta_E f(s_E(t)) + \sigma_E \xi_E(t) \quad (1)$$

$$\frac{dI(t)}{dt} = -\alpha_I I(t) + (1 - I(t))\beta_I f(s_I(t)) + \sigma_I \xi_I(t) \quad (2)$$

$$s_E(t) = W_{EE}E(t) - W_{EI}I(t) + h_E \quad (3)$$

$$s_I(t) = W_{IE}E(t) - W_{II}I(t) + h_I \quad (4)$$

The sigmoid response function is taken as $f(x) = (1 + e^{-x})^{-1}$. The deterministic Wilson-Cowan model ($\sigma_E = \sigma_I = 0$) can exhibit steady-state behavior (without oscillations) or coherent oscillations, depending on the parameter values. The transition between the two regimes is known as the Hopf bifurcation (see the black dotted line in Fig. 1-**(B)**) [28]. When noise is considered, the steady state behavior leads to noise-induced oscillations (left side of the black line in Fig1-**(B)**) while coherent oscillations become noise-perturbed (right side of the black line in Fig1-**(B)**). Our theory focuses on both noise-induced and noise-perturbed oscillations in the gamma band (30-100 Hz). However, most of the simulations shown in the main text are performed using noise-induced rhythms, since such oscillations are observed across brain areas in various species [5, 33]. We are interested in the coupling of gamma rhythms with a 10-Hz slow oscillation.

Various neural network architectures have been proposed to account for the emergence of cross-frequency coupling [14]. Here, we focus on the case in which a neural network generating fast rhythms is subject to slow exogenous inputs as sensory stimuli (Fig.1-**(B)**). Our goal is to understand the mechanisms by which cross-frequency coupling emerges and to propose a simplified, unified theory that accounts for the different signatures of CFC observed in data. In more detail, we aim to elucidate how network properties (intrinsic amplitude, frequency, stochasticity, dynamical regime, etc.) and parameters of slow external inputs (amplitude, frequency) interact to produce distinct signatures of cross-frequency neural oscillations.

2 Approximate dynamics of SWC LFPs subject to slow exogenous inputs: Parametric and Additive couplings

We model slow exogenous inputs as sinusoidal functions with constant amplitudes and fixed frequencies. For generality, we assume that excitatory and inhibitory neurons receive different inputs, $h_E^{ext}(t)$ and $h_I^{ext}(t)$, described as follows.

$$h_E^{ext}(t) = A_E^0 + A_E \sin(\omega_E t + \phi_E^0) \quad (5)$$
$$h_I^{ext}(t) = A_I^0 + A_I \sin(\omega_I t + \phi_I^0). \quad (6)$$

In these expressions, A_E^0 (respectively A_I^0) are the baselines, A_E (respectively A_I) the amplitudes, ω_E (respectively ω_I) the frequencies and ϕ_E^0 (respectively ϕ_I^0) the phases of the inputs applied to the E population (respectively the I population). Therefore, the total synaptic inputs applied to excitatory and inhibitory neurons in the network become

$$s_E(t) = W_{EE}E(t) - W_{EI}I(t) + h_E + h_E^{ext}(t) \quad (7)$$
$$s_I(t) = W_{IE}E(t) - W_{II}I(t) + h_I + h_I^{ext}(t) \quad (8)$$

To understand how the slow exogenous inputs Eqs. 5-6 interact with the fast rhythms, we perform a Taylor expansion of the former Stochastic Wilson Cowan dynamics subject to these inputs (Eqs. 1-2 and Eqs. 7-8). The idea is to assume that the meaningful quantities are the deviations from the baselines (fixed points of the WC dynamics without noise or exogenous inputs) of the activities of the Excitatory and Inhibitory populations (see Materials and Methods). These fluctuations are considered measures of Local Field potentials (LFPs). For simplicity, we retain only the lowest-order terms that reproduce the transition between the two dynamical regimes as in the former SWC dynamics. The following nonlinear and stochastic equations for the LFPs are obtained,

$$\begin{aligned}\frac{dV_E(t)}{dt} &= A_{EE}V_E(t) + A_{EI}V_I(t) + L_{EE}(V_E)^2(t) + L_{EI}V_E(t)V_I(t) + B_{1E}(V_E)^3(t) + B_{2E}V_E(V_I)^2(t) + B_{3E}V_I(V_E)^2(t)\\ &\quad + \underbrace{C_{E_0}A_E\sin(\omega_E t + \phi_E^0)} + \underline{C_{E_1}V_E(t)A_E\sin(\omega_E t + \phi_E^0) + C_{E_{3b}}W_{EE}(V_E)^2(t)A_E\sin(\omega_E t + \phi_E^0)}\\ &\quad + \underline{C_{E_{3c}}W_{EI}V_E(t)V_I(t)A_E\sin(\omega_E t + \phi_E^0)} + \sigma_E\xi_E(t), \quad (9)\\ \frac{dV_I(t)}{dt} &= A_{IE}V_E(t) + A_{II}V_I(t) + L_{II}(V_I)^2(t) + L_{IE}V_E(t)V_I(t) + B_{1I}(V_I)^3(t) + B_{2I}V_E(V_I)^2(t) + B_{3I}V_I(V_E)^2(t)\\ &\quad + \underbrace{C_{I_0}A_I\sin(\omega_I t + \phi_I^0)} + \underline{C_{I_1}V_I(t)A_I\sin(\omega_I t + \phi_I^0) + C_{I_{3b}}W_{II}(V_I)^2(t)A_I\sin(\omega_I t + \phi_I^0)}\\ &\quad + \underline{C_{I_{3c}}W_{IE}V_E(t)V_I(t)A_I\sin(\omega_I t + \phi_I^0)} + \sigma_I\xi_I(t), \quad (10)\end{aligned}$$

The above equations have complex conjugate eigenvalues defined as

$$\lambda = -\nu \pm j\omega_0 \quad (11)$$

The exact expressions of ν and ω_0 are obtained through linear stability analysis and depend on the state of the network. The interaction between slow exogenous inputs and fast gamma dynamics manifests in two distinct ways. We can identify multiplicative couplings (see terms underlined in Eqs. 9-10) and additive couplings (see terms underbraced in Eqs. 9-10). How multiplicative couplings shape the intrinsic dynamics? What is the role of the additive couplings on the overall rhythms? The answers to these questions are the motivations of our work. Since external inputs evolve on a slower timescale than the intrinsic dynamics, multiplicative couplings can be treated as constant over a period of the inherent oscillation. Over short intervals, multiplicative couplings function as parameters that modulate the dynamics of intrinsic rhythms according to their current values. Such couplings can increase or decrease the values of the inherent rhythm components (amplitude, frequency, and phase) depending on the time epoch. We call them parametric couplings. The additive couplings can be seen as external forces that tend to entrain the intrinsic fast dynamics. They clearly behave differently from multiplicative couplings. To disentangle the roles of these two interactions in the PAC phenomenon, we will treat them separately. For the rest of our work, we will use the Approximate SWC.

3 Parametric couplings are responsible for the PAC mechanism

To identify the roles of the parametric and the additive couplings, respectively, we investigate their effects separately. We begin by examining the role of parametric couplings. We consider Eqs. 9-10 without additive couplings (this means that we discard the underbraced terms in Eqs.9-10). We look for their dynamics in the forms.

$$V_{E_f}^{par}(t) = Z\cos\left(\omega_0 t + \phi\right) \tag{12}$$
$$V_{I_f}^{par}(t) = \alpha Z\cos\left(\omega_0 t + \phi + \delta\right) \tag{13}$$

Where, Z and ϕ are the envelope and phase dynamics, $\omega_0 = 2\pi f_0$ with f_0 the frequency and, α and δ two coefficients (see Linear stability analysis) which depend on the fast network parameters and the baseline inputs (A_E^0,A_I^0). It is essential to mention that the baseline inputs change the state of the intrinsic network. In other words, if the state of the network before stimulation is where the magenta point is on the left side of the bifurcation curve (Fig.1.**(B)**), the baseline inputs will shift it. The baseline inputs can also be seen as static stimuli. Therefore, the external inputs may be interpreted as the sum of static and time-varying inputs. In this work, we will focus on the effects of the time-varying part of the inputs. The theoretical results are obtained with nonzero baselines, whereas all simulations are performed with null baselines. The SAM allows a derivation of the envelope and phase dynamics as follows.

$$dZ(t) = \left[-\left(\nu + C_1^E A_E \sin(\omega_E t + \phi_E^0) + C_1^I A_I \sin(\omega_I t + \phi_I^0)\right)Z(t) + B_1 Z^3(t) + \frac{D}{2Z(t)}\right]dt + \sqrt{D}dW_1(t) \tag{14}$$
$$d\phi(t) = \left[C_2^E A_E \sin(\omega_E t + \phi_E^0) + C_2^I A_I \sin(\omega_I t + \phi_I^0) + B_2 Z^2(t)\right]dt + \frac{\sqrt{D}}{Z(t)}dW_2(t). \tag{15}$$

The parameters C_1^E, C_1^I, C_2^E, and C_2^I can be interpreted as the coupling coefficients of the fast intrinsic rhythms with the external inputs. Their values are usually small (less than 1), resulting in effective amplitudes of the exogenous inputs within the physiological range (approximately equal to the intrinsic amplitude of the rhythms). The coefficients $-\nu$ are the genuine parts of the complex conjugate eigenvalues of the fast network. The constants B_1 and B_2 account for the nonlinearity in the dynamics and allow the envelope-phase equations to describe the rhythms across the bifurcation (black dot line Fig.1-**(B)**) [23]. Without them, the envelope-phase dynamics will only work in the noise-induced regime [22]. The values of all these parameters depend on network state, which means the baseline inputs A_E^0 and A_I^0, and all the other network parameters. Our theory Eqs. 14-15 shows that the amplitude and phase dynamics of the intrinsic gamma rhythms depend both on the time-varying external inputs. We can visualize how the amplitude dynamics $Z(t)$ of the fast gamma rhythms (Eq.14) are shaped by the slow external stimulus (its amplitude, phase, and frequency). The same observation can be made for the phase dynamics $\phi(t)$ (Eq.14). The envelope and phase dynamics of the fast rhythms depend explicitly on the components of the slow rhythms; thus, on the Cross-Frequency Coupling mechanisms. Our envelope-phase equations can describe all four signatures of CFC, since all coupling mechanisms are present. In addition, in general, when the inputs to the E and I populations differ, our equations show that multiplexing CFC occurs. Our equations also describe the multiplexing case, in which several inputs are sent simultaneously to the E. I populations, in that case the coefficients C_1^i (i=E,I) will be equal, similarly for the coefficients C_2^i (i=E,I). Moreover, the reduced dynamics Eqs. 14-15 obtained through the envelope-phase decomposition is an excellent approximation of the LFPs dynamics Eqs.9-10 driven by sensory inputs without additive contributions, as we will show through numerical simulations. In this work, we focus on the Phase-Amplitude (PAC) and Phase-Frequency coupling mechanisms; however, we briefly discuss the Amplitude-Amplitude (AAC) and Phase-Phase (PPC) coupling behaviors.

3.1 PAC depends on the state of the fast network and the parameters of the inputs

To understand how PAC appears, let us consider the amplitude dynamics of the fast rhythms Eqs.14, which are coupled with the external inputs but uncoupled with the phase (Eqs.15) of the rapid rhythms. Without external inputs, the dynamics of the amplitude depend only on the fundamental part of the complex conjugate eigenvalues $-\nu$, D, and B_1 (see [23]). The value of $-\nu$ reflects the levels of synchrony and amplitude of the rhythms. In fact, the

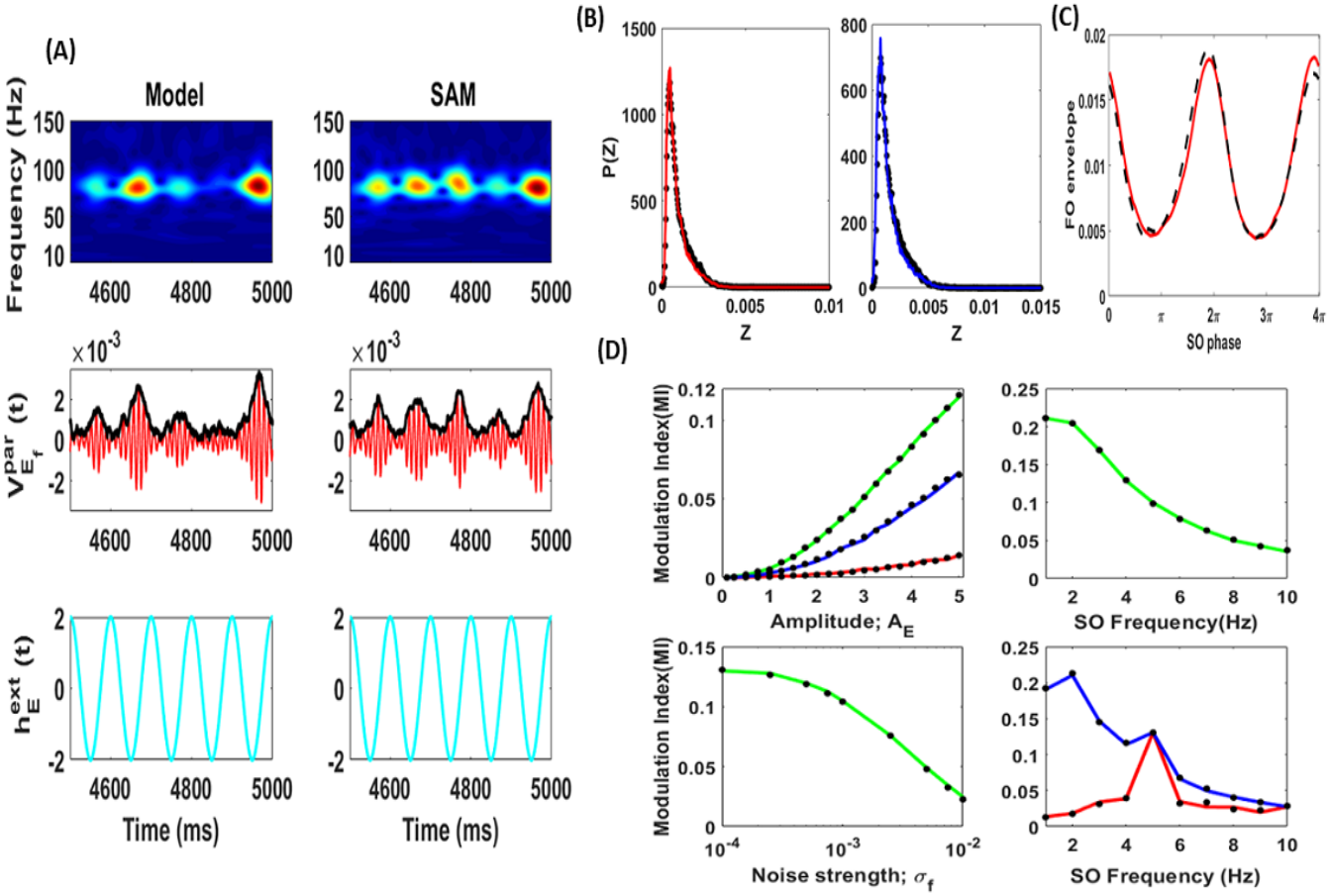

Figure 2. Parametric coupling of fast noise-induced gamma rhythms by slow sensory inputs applied on both the E and I populations. (A): Time frequencies (top) and time series (middle) of the E fluctuations subject to slow sensory inputs (bottom) applied to both E and I populations. The left panels correspond to the model, and the right panels are obtained from the SAM analysis. The envelope dynamics (black dynamics in the middle panels) are obtained from the Hilbert transform for the model and from Eqs. 14-15 for the SAM. Visual inspection reveals modulation of the envelope of the fast rhythms (black dynamics in the middle panels) by the phase of the slow external stimulus (bottom panels). **(B):** Probability densities of the envelope dynamics extracted through the Hilbert transform (colored curves) and through the SAM dynamics (black dots). The left panel corresponds to the E dynamics, whereas the right panel corresponds to the I dynamics. **(C):** Modulation density of the fast oscillation envelope of the E fluctuations with the phase of the slow phase of the external input. The red curve corresponds to the model, and the dashed black line is obtained from the SAM envelope. **(D):** Modulation properties of the envelope of the E dynamics by the phase of the slow external input. In the top left panel, an increase in the modulation index as the amplitude of the external input increases. Green, red, and blue curves correspond respectively to the cases where the external inputs are applied to both E and I dynamics, the I dynamics alone, and E dynamics alone. In the top-right panel, a decrease in the modulation index is observed as a function of the frequency of the slow external input. Bottom-left, decrease in the modulation index with the intensity of the noise in the fast dynamics. Bottom-right, variation of the modulation index when the frequency input to the I population varies (blue curve) while the input to the E population remains the same (red curve). For all panels, black dots correspond to the SAM dynamics Eqs.14-15 and the parameters are: $\alpha_E = 0.1$, $\alpha_I = 0.2$, $\beta_E = 1.0$, $\beta_I = 2.0$, $W_{IE} = 26.3$, $W_{EI} = 32$, $W_{II} = 1.3$, $h_E = -3.8$,$h_I = -8.0$, $W_{EE} = 27.4$, $A_E = 2$,$f = 10Hz$,$\phi_E^0 = \phi_I^0 = \pi/2$. For panels **(A-C)**, the noise intensity is set to $\sigma = 0.0001$. For panels **D**, the frequency of the slow external input has been set to $f = 5Hz$ and its amplitude to $A_E = 3$ except the first panel (left-Top) where the external frequency has been kept at $f = 10Hz$. The noise intensity is $\sigma = 0.0001$.

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amplitude of the fast rhythms (see Fig.1 increases as $-\nu$ does. When the slow external inputs are considered, we can define an effective fundamental part $-\nu_{ef}$ of the complex conjugate eigenvalues from

$$\nu_{ef}(t) = \nu + C_1^E A_E \sin(\omega_E t + \phi_E^0) + C_1^I A_I \sin(\omega_I t + \phi_I^0) \tag{16}$$

For clarity, we restrict our analysis to the case where the external inputs to the E and I populations are identical ($A_E = A_I = A; \omega_E = \omega_I = \omega; \phi_E^0 = \phi_I^0 = \phi_0$). The expression of ν_{ef} is reduced to

$$\nu_{ef}(t) = \nu + (C_1^E + C_1^I) A \sin(\omega t + \phi_0) \tag{17}$$

The effective real part of the complex-conjugate eigenvalue becomes time-dependent and periodic, with the same frequency as the slow external input. Since $-1 \leq \sin(\omega t + \phi_0) \leq 1$, the value of $\nu_{ef}(t)$ oscillates between a maximum $\nu + (C_1^E + C_1^I)A$ when $\sin(\omega t + \phi_0) = 1$ or $\omega t + \phi_0 = \frac{\pi}{2}$ and a minimum $\nu - (C_1^E + C_1^I)A$ when $\sin(\omega t + \phi_0) = -1$ or $\omega t + \phi_0 = -\frac{\pi}{2}$. The amplitude Z of the fast rhythms is modulated by the phase $\omega t + \phi_0 = \frac{\pi}{2}$ of the slow external input. Therefore, the effective real part is a periodic function that obeys the following relation

$$-\nu - (C_1^E + C_1^I)A \leq -\nu_{ef}(t) \leq -\nu + (C_1^E + C_1^I)A. \tag{18}$$

This implies Phase-Amplitude coupling (PAC). Moreover, numerical simulations of the system of equations 9-10 without additive couplings (we consider this as our Model) show excellent agreement with the envelope-phase decomposition Eqs.14-15. In our simulations, we consider the 'Model' to be the LFP dynamics Eqs. 9-10, and the 'SAM' refers to the dynamics Eqs. 12-15. We further consider cases in which the fast network, before interacting with the slow external inputs, is in the noise-induced or noise-perturbed regime.

3.1.1 PAC in the noise-induced regime

The behavior of PAC in the noise-induced regime (see magenta dot, left side of the bifurcation Fig.1.**(B)**) is reported in Figs.2. We see that values of the fast gamma envelope (black curves, middle panels) are higher at a specific phase of the slow input (cyan curves, bottom panels). We also observe a good matching between the model Eqs.9-10 (without additive contribution) and its corresponding envelope-phase dynamics (Eqs.14-15) obtained through the SAM. Before coupling with external inputs, the noise-induced regime exhibits bursts with random durations, amplitudes, and appearances [22, 33]. We observe that slow external inputs shape the burst duration, appearance, and amplitude. In fact, bursts occur primarily during a particular phase of the external inputs. Their duration and amplitude also seem to be correlated with the period and the amplitude of the slow inputs. However, a complete analysis of burst dynamics and their relationship with the input is outside the scope of this work. We further compare the envelope probability density functions of the model (E and I) obtained via the Hilbert transform and the SAM in Fig.2(B) and find good agreement.

The PAC strength can be quantified by computing the probability distribution of the fast signal's envelope as a function of the phase of the slow external input. Specifically, for each phase of the external input, we determine the corresponding envelope of the fast gamma rhythms and the associated mean. The probability that the envelope values correspond to a specific phase of the external input can then be computed. The results of this approach are shown in Figs.2(C), where we can see that peaks of the amplitude probability are located at a specific phase of the slow external input. This implies Phase-amplitude coupling. Furthermore, we observe a good match between the SAM (dashed black line) and the envelope obtained through the Hilbert transform (red solid line). A single value can also quantify the PAC, the modulation index (MI). We are specifically interested in the modulation properties as we vary different parameters of the slow external signals (amplitude, frequency) and the fast gamma rhythms (noise intensity). For simplicity, we first consider the case where the exogenous inputs to the E and I neurons are identical $\omega_E = \omega_I = \omega$. We keep the frequency at $f = 10Hz$ and increase the amplitude A_E, which is also set identically for both populations. We found that the MI increases with the amplitude. If the input is applied only on one of the two populations (E or I, see blue and red curves Fig.2-**(D)**, first panel) the corresponding MI is lower than if it was applied on both populations (see green curve Fig.2-**(D)**, first panel). Moreover, the I population produces a stronger modulation than the E population. In contrast to the amplitude, increasing the frequency decreases the MI (see colored curves Fig.2-**(D)**, second and last panels). When the noise intensity in the fast rhythm is increased, the MI vanishes. Finally, we computed the MI when the inputs applied to the two populations differ (Fig.2-**(D)**-Bottom,

Right). Specifically, we keep the input fixed at the E population and increase the input frequency to the I populations. The MI for the I populations decreases, whereas that for the E populations remains almost constant, except when the frequencies of both populations are identical. Overall, the mechanisms of Phase-Amplitude coupling in the noise-induced regime depend on both input and intrinsic network parameters.

3.1.2 PAC in the noise-perturbed regime

We shift to the noise-perturbed regime (Fig.1-**(B)**, right side of the bifurcation). In this regime, oscillations persist in the absence of noise and do not exhibit bursts before coupling to external inputs. The envelopes of the fast rhythms (Fig.3-**(A)**, black curves, middle panels) are almost periodic and correlated with the external inputs in both the Model and the SAM. In fact, their peaks occur at a specific phase of the external input. In contrast to the noise-induced regime, the envelopes of the fast rhythms seem more regular and perfectly periodic. In addition, their probability densities are no longer unimodal as in the noise-induced regime, but bimodal. The transition to the noise-perturbed regime has induced a second peak in the envelope PDF. This is an important biomarker that could enable discrimination between PAC in the noise-induced and noise-perturbed regimes in real data. We also computed the probability of the fast-rhythm envelope as a function of the phase of the slow input (Fig.3-**(C)**) and confirmed the presence of PAC. Finally, we computed the MI and found properties similar to those observed in the noise-induced regimes. The critical difference between the two regimes is the bimodal nature of the PDF in the noise-perturbed regime. Our work complements previous results [24], in which PAC was investigated in the Stuart-Landau model. Our analysis here extends prior work by considering the two oscillatory regimes.

4 Additive couplings can be interpreted as a time-varying baseline

We now consider the additive terms. They are considered periodic external forces. In other words, they tend to lock the dynamics of the fast rhythms to their own dynamics (which are slow). We assume trials solutions for the additive terms only (Eqs.9-10), in the following forms

$$V_{E_s}^{add}(t) = a_1 \cos(\omega_E t) + b_1 \sin(\omega_E t) + c_1 \cos(\omega_I t) + d_1 \sin(\omega_I t) \tag{19}$$

$$V_{I_s}^{add}(t) = a_2 \cos(\omega_E t) + b_2 \sin(\omega_E t) + c_2 \cos(\omega_I t) + d_2 \sin(\omega_I t) \tag{20}$$

To obtain the expressions of the coefficients a_1, b_1,c_1,d_1, a_2, b_2,c_2 and d_2, we consider the deterministic dynamics ($\sigma_E = \sigma_I = 0$) of Eqs. 9-10 without parametric terms (see terms underline in Eqs. 9-10). For simplicity and to obtain approximate forms of these coefficients, we discard all nonlinear terms. This additional simplification does not mislead the solutions, particularly for noise-induced oscillations and noise-perturbed solutions near the bifurcation. We replace these trial solutions in the deterministic, linear version of Eqs. 9-10 and solve the system of equations 52 for a_1, b_1,c_1,d_1, a_2, b_2,c_2 and d_2 (see Eqs. 52). The solutions with only additive terms oscillate at the external frequencies. They are slow inputs for the choice of our parameters. We found that they do not induce any additive modulation of the amplitude or phase of the fast rhythms. They act as a time-varying baseline for the fast envelope-phase dynamics Eqs.14-15. Therefore, a full solution of the approximate dynamics Eqs.9-4 is chosen as the superposition of both parametric and additive contributions of the external inputs:

$$V_E(t) = V_{E_f}^{par}(t) + V_{E_s}^{add}(t) = Z\cos\left(\omega_0 t + \phi\right) + a_1 \cos(\omega_E t) + b_1 \sin(\omega_E t) + c_1 \cos(\omega_I t) + d_1 \sin(\omega_I t) \tag{21}$$

$$V_I(t) = V_{I_f}^{par}(t) + V_{I_s}^{add}(t) = \alpha Z\cos\left(\omega_0 t + \phi + \delta\right) + a_2 \cos(\omega_E t) + b_2 \sin(\omega_E t) + c_2 \cos(\omega_I t) + d_2 \sin(\omega_I t). \tag{22}$$

If the network is driven by sensory inputs at the same frequency ω, the solution above is reduced to

$$V_E(t) = V_{E_f}^{par}(t) + V_{E_s}^{add}(t) = Z\cos\left(\omega_0 t + \phi\right) + a_1 \cos(\omega t) + b_1 \sin(\omega t) \tag{23}$$

$$V_I(t) = V_{I_f}^{par}(t) + V_{I_s}^{add}(t) = \alpha Z\cos\left(\omega_0 t + \phi + \delta\right) + a_2 \cos(\omega t) + b_2 \sin(\omega t). \tag{24}$$

The closed forms of the coefficients can be obtained through the Eqs. 54-57. Using the above theoretical expressions, we perform numerical simulations and compare the results with those obtained from the approximate SWC Eqs.9-10 and the former SWC Eqs.1-8. We found good agreement between the three dynamics as shown in

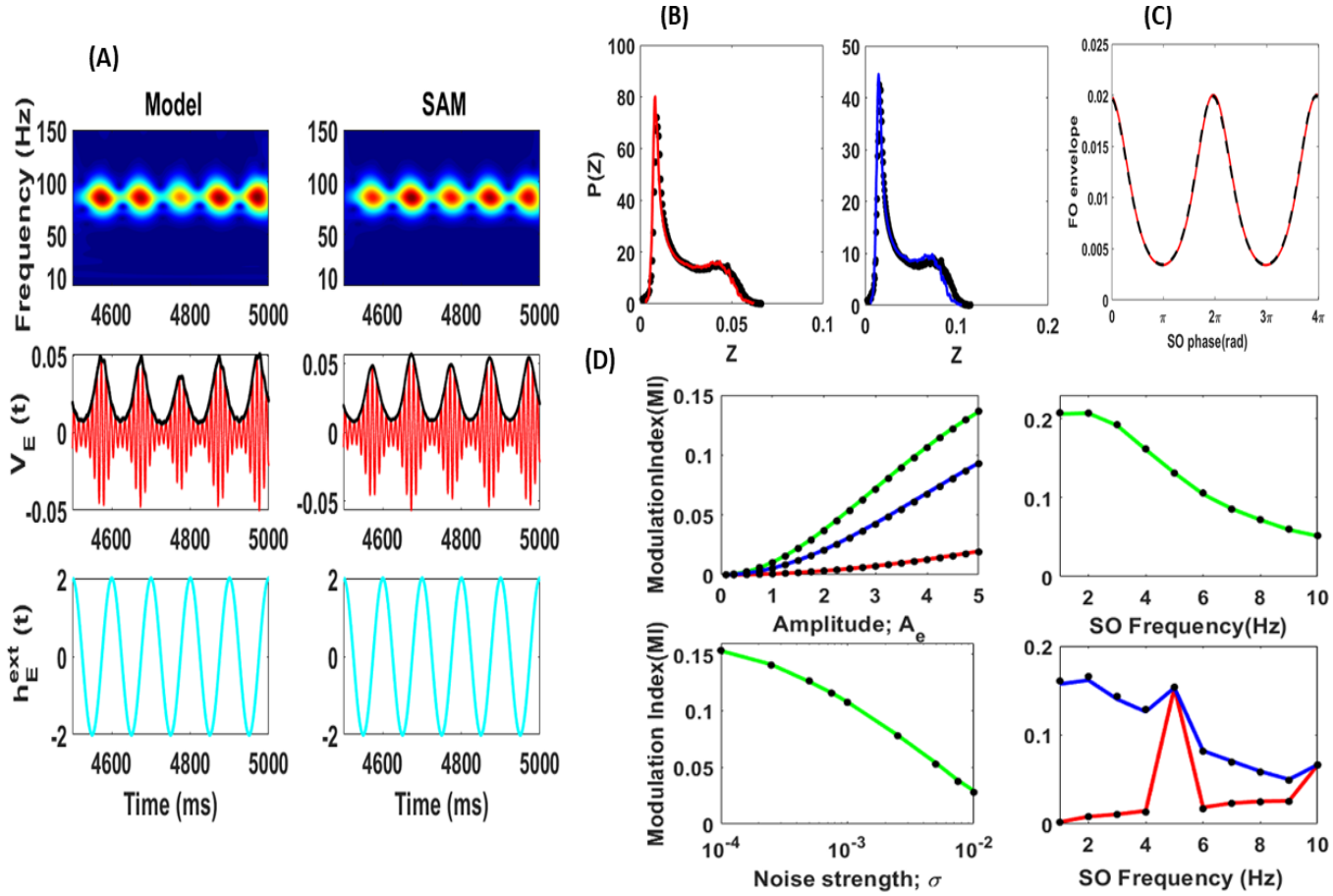

Figure 3. Parametric modulation of fast noise-perturbed gamma rhythms by slow sensory inputs applied on both the E and I populations. (A): Time frequencies (top) and time series (middle) of the E fluctuations subject to slow sensory inputs (bottom) applied to both E and I populations. The left panels correspond to the model, and the right panels are obtained from the SAM analysis. The envelope dynamics (black dynamics in the middle panels) are obtained from the Hilbert transform for the model and from Eqs. 14-15 for the SAM. Visual inspection reveals modulation of the envelope of the fast rhythms (black dynamics in the middle panels) by the phase of the slow external stimulus (bottom panels). **(B):** Probability densities of the envelope dynamics extracted through the Hilbert transform (colored curves) and through the SAM dynamics (black dots). The left panel corresponds to the E dynamics, whereas the right panel corresponds to the I dynamics. **(C):** Modulation density of the fast oscillation envelope of the E fluctuations with the phase of the slow phase of the external input. The red curve corresponds to the model, and the dashed black line is obtained from the SAM envelope. **(D):** Modulation properties of the envelope of the E dynamics by the phase of the slow external input. Top left panel, decrease in the modulation index as a function of the amplitude of the external input. The green curve corresponds to the case where the external inputs is applied to both E and I dynamics, the blue curve is obtained when the external inputs targets only I population and the red curve is the case when the input is applied only to the E population. Top right panel, decrease in the modulation index as a function of the frequency of the slow external input. Bottom-left, decrease in the modulation index with the intensity of the noise in the fast dynamics. Bottom-right, variation of the modulation index when the frequency input to the I population varies (blue curve) while the input to the E population remains the same (red curve). For all panels, black dots correspond to the SAM dynamics Eqs.14-15.

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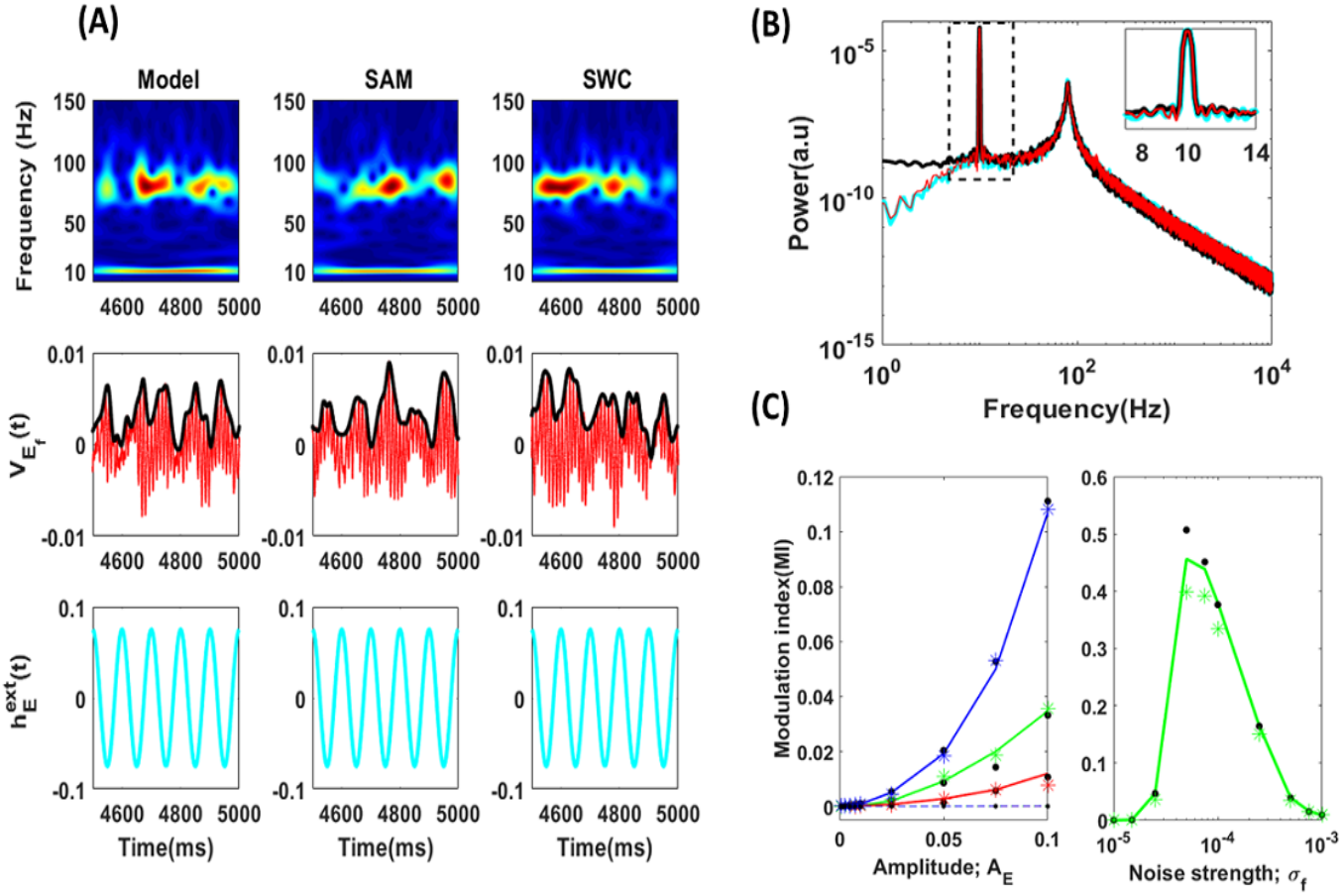

Figure 4. Total modulation of fast noise-induced gamma rhythms by slow sensory inputs applied on both the E and I populations. (A): Time frequencies (top) and time series (middle) of the E fluctuations subject to slow sensory inputs (bottom) applied to both E and I populations. The left panels correspond to the model, the bottom panels match the SAM analysis, and the right panels are obtained using the full SWC model. The envelope dynamics (black dynamics in the middle panels) are obtained after interpolating the local maxima of the rhythms. Visual inspection reveals modulation of the envelope of the fast rhythms (black dynamics in the middle panels) by the phase of the slow external stimulus (bottom panels). **(B):** Power spectra of the E fast dynamics obtained from the model (cyan), the SWC (red), and through the SAM dynamics (black). The dashed rectangle shows pics at 10 Hz as signatures of the slow sensory inputs. The inset shows a match between the Model, the SWC, and the SAM. **(C):** Modulation Index (MI) of the fast oscillation envelopes of the E fluctuations by the phases of the slow phases of the external input. The left panel shows the increase of the MI as a function of the amplitude A_E of the sensory input for the total modulation. Solid-colored lines correspond to the model, crosses denote outcomes of the SWC dynamics, and black dots denote results from the SAM analysis. The red color corresponds to the case where the input is applied to the E population alone. The blue color indicates that the input is applied to the I population alone, and the green color indicates that the input is applied to both the E and I populations. Note that modulation happens at very low values of the amplitude of the external input compared to the pure modulation case (dashed blue line and the corresponding black dots for the SAM). The right panel shows the non-monotonic variation of MI with increasing noise strength. The solid green line corresponds to the model, the crosses to the SWC, and the black dots to the SAM. For all the panels the parameters are: $\alpha_E = 0.1$, $\alpha_I = 0.2$, $\beta_E = 1.0$, $\beta_I = 2.0$, $W_{IE} = 26.3$, $W_{EI} = 32$, $W_{II} = 1.3$, $h_E = -3.8$,$h_I = -8.0$, $W_{EE} = 27.4$,$f = 10Hz$,$\phi_E^0 = \phi_I^0 = \pi/2$. For panels **(A,B)** and **(C)**-Right, the external input intensity is set to $A_e = 0.075, 0.5$ and 0.1 respectively.

Fig.4. The time-frequency representations in the top panels highlight the signatures of both the fast gamma rhythms and the slow exogenous inputs. This is also evident in the power spectra (Fig.4-**(A)**), where good agreement among the three approaches is observed.

Our theory is robust in the noise-induced regime and for low values of the slow external input amplitudes. In the noise-perturbed regime, we still observe good agreement near the bifurcation. However, going far from the bifurcation leads to divergence between the two approximations (SAM Eqs.21-22 and approximated SWC Eqs.9-10) and the former SWC.

4.1 LFPs of the SWC driven by exogenous inputs can be described as the sum of additive and parametric contributions

Using the Taylor expansion and the Stochastic Averaging Method (SAM), we have shown that SWC dynamics driven by exogenous inputs can be described as the sum of parametric terms, which account for PAC, and additive terms, which describe the frequency-locking of the fast dynamics to the inputs. However, the question of directly extracting these parametric and additive couplings from the time series of SWC dynamics remains unresolved. In the context of neural data, imagine that you record raw data from the hippocampus, for example, and you have serious evidence that such data are cross-frequency coupled, how could you extract their parametric and additive contributions? To answer these questions, we interpret our reduced dynamics obtained from the SAM. The parametric contributions described the fast rhythms modulated by the phase of the slow exogenous inputs. Therefore, information about the parametric contributions is contained in the fast dynamics. An appropriate way will be to bandpass filter the LFPs generated by the SWC Eqs.1-2 and Eqs.7-8 in the gamma band to extract only its parametric contribution. Similarly, to extract the additive contributions, filtering around the exogenous input frequency is advisable. The bandpass filtering method is widely used in neuroscience, where raw brain data are typically filtered to the frequency band of interest before analysis. This is usually the case for the study of cross-frequency amplitude coupling [12, 29]. We then simulated the SWC driven by exogenous inputs and filtered the corresponding time series (raw data) in the gamma and alpha frequency bands. The corresponding filtered signal in the gamma band is analysed and compared to the parametric contributions obtained from the SAM. Similarly, the filtered signal in the alpha band is compared to the additive contributions.

The results of this approach are shown in Figs.5 where we have represented the LFPs obtained from numerical simulations (Figs.5-**(A-B)**, top panels), the corresponding parametric (Figs.5-**(A-B)**, middle panels) and additive (Figs.5-**(A-B)**, bottom panels) contributions for low and high values of the exogenous inputs amplitude, respectively. For the SAM, we represented respectively the sum (top panels) of the parametric (midlle panels) and additive (bottom panels) contributions. We further compute the modulation Index (MI) as the amplitude and the frequency of the exogenous inputs increase. We found a good match between the filtered LFPs obtained from the SWC timeseries and the theoretical expressions from the SAM. The discrepancy between the two values was due to the filtering method employed. In fact, we found that MI outcomes were sensitive to the filtering parameter. However, the qualitative behavior of the MI was the same for both the filtered LFP and envelope-phase decomposition. This is confirmed by the decrease in MI as noise in the fast rhythms increases.

The behavior of the LFPs and their corresponding MI depend on the parameters of both the exogenous inputs and the fast intrinsic dynamics, as described in the SAM solutions. Our theory shows that CFC can be defined as the superposition of parametric and additive couplings, which respectively account for the fast and slow components of the coupling. It provides a biologically realistic theoretical framework for understanding cross-frequency coupling in general and phase-amplitude coupling (PAC) in particular. The exogenous inputs were assumed to be sensory inputs and modelled as coherent sinusoidal functions. However, high-speed networks are often coupled to other networks oscillating at lower frequencies. In the following, we describe such situations from the approach and results obtained here.

Discussion and Conclusion

We have proposed a simple theory of coupled oscillations, each evolving on different timescales. Our theory describes a system of coupled fast-slow rhythms as the sum of two sinusoidal functions. One of the two functions named here as the parametric coupling represents the modulation of the envelope of fast rhythms by the phase of the slow one, or

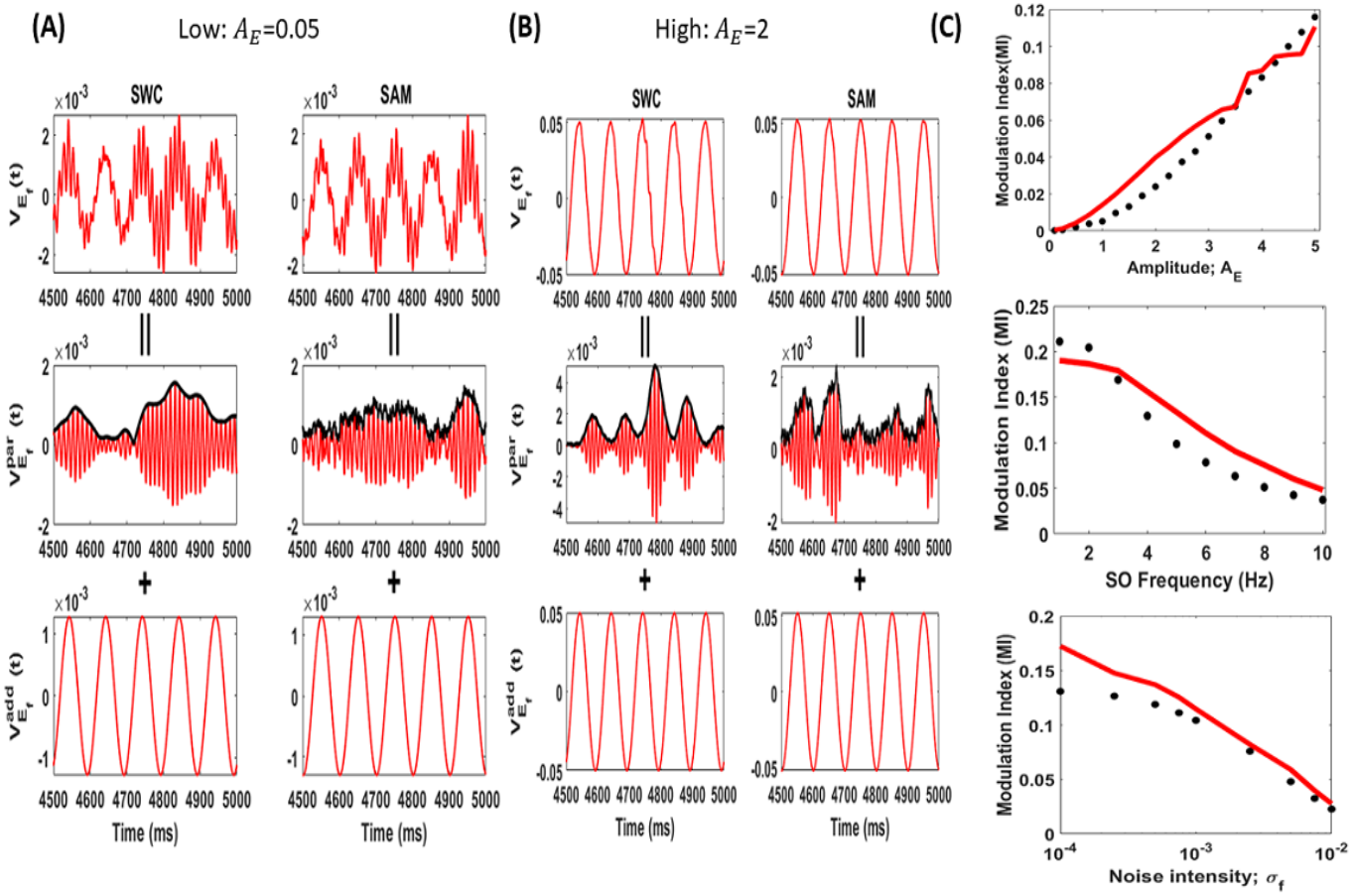

Figure 5. Decomposition of the SWC LFP timeseries into their parametric and additive contributions. **(A)-(B)**: SWC LFPs obtained after numerical simulations (Top panels) of Eqs.1,2,7 and 8 are filtered in the gamma band (middle panels) to obtain parametric contributions responsible for the PAC, and in the alpha band (bottom panels) to obtain additive contributions at the origin of the entrainment. For the SAM, LFPs (top panels) are the sum of parametric and additive contributions. **(A)**: The value of the exogenous input amplitude is low ($A_E = 0.05$).**(B)**: The value of the exogenous input is high ($A_E = 2$). We computed the Modulation Index (MI) as the amplitude (**(C)**-top), the frequency (*(C)*-middle) of the exogenous inputs is increased. We also computed the MI as the noise in the fast network decreases. For all figures required, the noise intensity in the fast network is set to $\sigma = 0.0001$. In panel **(C)-top**, the frequency is set to $f = 10Hz$, in panel **(C)-middle** the amplitude is $A_E = 2$ and in panel **(C)-bottom**, the frequency is $f = 5Hz$ and the amplitude $A_E = 2$. For all panels, other parameters are specified in the previous figures.

PAC. A recent study has also described the PAC mechanism by a similar parametric coupling [24]. The results obtained here, showing the increase in the MI as the amplitude of the external input increases or its decrease as the frequency (of the external input) increases, are in good agreement with that study. However, those (in [24]) were obtained in the limit-cycle regime without noise. Moreover, the model employed was the canonical Stuart-Landau model, which lacks explicit biological parameters, although it is widely used to simulate brain rhythms. On the contrary, our approach starts from the Wilson-Cowan dynamics, which are recognized to account for the biological parameters of a neural population. We incorporate noise, another essential ingredient of gamma rhythms seen in vivo. This allows us to describe both noise-induced and noise-perturbed rhythms, and represents a new modelling approach. The parametric coupling described by the SAM allows a clear visualisation of the coupling between the envelope of the fast rhythms and the phase of the slow rhythms, thereby elucidating the PAC mechanism. Another difference between our results and those in [24] is that we extend the description of the PAC mechanism by incorporating additive coupling. This latter coupling enables us to obtain a complete description of the LFPs in terms of stochastic Wilson-Cowan dynamics. The results of this modelling are in agreement with previous modelling of CFC LFPs [29]. Then, CFC can be described as a superposition of parametric and additive couplings between the fast and slow rhythms. This description applies to the two principal architectures: a network that generates fast oscillations and is externally driven by slow sensory inputs, and a network that is coupled to another network oscillating at a low frequency. For this latter architecture, we show that our theory reproduces the phenomenon of frequency-dependent interaction, which has been widely observed in recordings of humans and non-human primates [3, 19, 30]. In fact, the phenomenon suggests that the system of fast-slow networks coupled in a feedback-feedforward configuration shares information using specific frequency channels [18, 21]. This is evidenced by coherence, Granger causality, and the directed asymmetry index (DAI), with good agreement between our theories and the results of direct numerical simulations of the stochastic Wilson-Cowan model. We use our theory to investigate the mechanism of long-distance communication between brain areas coordinated by slow oscillations [7, 8, 10, 31]. The results on coherence between fast local oscillations across regions indicate that communication can occur via slow oscillations, consistent with biological predictions. Our theory, therefore, reproduces some of the essential behaviors and functions of the CFC. We believe our work is an important step towards understanding CFC.

Future investigations should use our formulation to study other CFC signatures (PPC, PFC, AAC) [14]. Long-distance communication coordinated by coherent theta oscillations in the hippocampus has been reported in several studies. In our work, we investigated only the noise-induced case; it is therefore essential to also examine the more plausible case in which they are coherent. Finally, we hope our framework will serve as a basis for more specialized studies on the role and functions of CFC in hippocampal learning and memory.

Materials and Methods

Dynamics of the Stochastic Wilson Cowan

We consider the Stochastic Wilson Cowan (SWC) model of neural rhythms driven by slow exogenous inputs. The equations are described as follows.

$$\begin{aligned}
\frac{dE(t)}{dt} &= -\alpha_E E(t) + (1-E(t))\beta_E f(s_E(t)) + \sigma_E \xi_E(t) \\
\frac{dI(t)}{dt} &= -\alpha_I I(t) + (1-I(t))\beta_I f(s_I(t)) + \sigma_I \xi_I(t) \\
s_E(t) &= W_{EE}E(t) - W_{EI}I(t) + h_E + h_E^{ext}(t) \\
s_I(t) &= W_{IE}E(t) - W_{II}I(t) + h_I + h_I^{ext}(t)
\end{aligned} \tag{25}$$

Without any restriction, we defined the two exogenous inputs to the E and I cells as

$$h_E^{ext}(t) = A_E^0 + A_E \sin(\omega_E t + \phi_E^0) \tag{26}$$

$$h_I^{ext}(t) = A_I^0 + A_I \sin(\omega_I t + \phi_I^0) \tag{27}$$

The parameters A_E^0 and A_I^0 are the baselines of the signals h_E^{ext} and h_I^{ext}, A_E and A_I are their amplitudes and, ϕ_E^0 and ϕ_I^0 the initial phases. The external sensory inputs h_E^{ext} and h_I^{ext} oscillate at low frequencies $]0, 10Hz]$ ω_E and ω_I respectively. The two independent Gaussian white noises have the following properties

$$\begin{aligned}
\left\langle \xi_{E,I}(t) \right\rangle &= 0, \quad \left\langle \xi_E(t)\xi_I(t) \right\rangle = 0 \\
\left\langle \xi_E(t')\xi_E(t) \right\rangle &= \left\langle \xi_E(t')\xi_E(t) \right\rangle = \delta(t-t')\,.
\end{aligned}$$

The sigmoid response function is taken as $f(x) = (1+e^{-x})^{-1}$. The model without exogenous inputs can generate oscillations across different frequency bands, depending on the specific parameter choices. Also, the rhythms can be noise-perturbed or noise-induced. In this work, the mean frequency of the oscillations is set to belong to the gamma band ([30 100] Hz). The coefficient W_{EE} is chosen to modify the dynamical regime of the rhythms (noise-perturbed or noise-induced).

Dynamics of the fluctuations: The SWC3 model

To have a better understanding of the role of the sensory inputs h_E^{ext} and h_I^{ext}, on the SWC dynamics. We consider the set of parameters for which the model without noise and sensory inputs has a single fixed point (E_0, I_0) and complex-conjugate eigenvalues ($\lambda = -\nu \pm j\omega_0$). This means that the model without sensory inputs exhibits oscillations. We recall that these oscillations can be subdivided into two distinct types: Noise-induced, also called quasi-cycle rhythms, which arise from Gaussian white noise and are observed when the fundamental part of the complex conjugate eigenvalues is negative. Noise perturbed or noisy limit cycle oscillations, which are seen when the fundamental part of the complex conjugate eigenvalue is positive. When noises and the sensory inputs are included, the noise-free dynamics are perturbed. We are interested in the deviations from the fixed points. The E and I activities can be separated as follows:

$$E(t) = E_0 + V_E(t) \qquad and \qquad I(t) = I_0 + V_I(t) \tag{28}$$

where the fixed points E_0 and I_0 solve the deterministic equations:

$$-\alpha_E E_0 + \big(1-E_0\big)\beta_E f(s_{E_0}) = 0 \tag{29}$$

$$-\alpha_I I_0 + \big(1-I_0\big)\beta_I f(s_{I_0}) = 0\,, \tag{30}$$

with the definitions

$$\begin{aligned} s_{E_0} &= W_{EE}E_0 - W_{EI}I_0 + h_E + A_E^0 \\ s_{I_0} &= W_{IE}E_0 - W_{II}I_0 + h_I + A_I^0 \,. \end{aligned}$$

It appears that the baseline coefficients A_E^0 and A_I^0 modify the state of the network by modifying the values of the fixed points E_0 and I_0. Using Eqs. (28) The Taylor expansion of the sigmoid function to the second order can be obtained as

$$\begin{aligned} f(s_E(t)) &= f(s_{E_0} + \delta s_E(t)) \\ &= f(s_{E_0}) + \delta s_E(t) f^{'}(s_{E_0}) + \frac{1}{2}(\delta s_E(t))^2 f^{''}(s_{E_0}) \\ f(s_I(t)) &= f(s_{I_0} + \delta s_I(t)) \\ &= f(s_{I_0}) + \delta s_I(t) f^{'}(s_{I_0}) + \frac{1}{2}(\delta s_I(t))^2 f^{''}(s_{I_0}) \end{aligned} \tag{31}$$

where

$$\begin{aligned} \delta s_E(t) &= W_{EE}V_E(t) - W_{EI}V_I(t) + A_E \sin(\omega_E t + \phi_E^0) \\ \delta s_I(t) &= W_{IE}V_E(t) - W_{II}V_I(t) + A_I \sin(\omega_I t + \phi_I^0). \end{aligned} \tag{32}$$

Replacing the Taylor expansion of the sigmoid function in the Wilson-Cowan dynamics and taking into account the fixed point conditions, we arrive at the following approximate equations:

$$\begin{aligned} \frac{dV_E}{dt} &= -\alpha_E V_E + (1-E_0)\beta_E f^{'}(s_{E_0})\delta s_E - \beta_E f(s_{E_0})V_E + \underbrace{(1-E_0)\beta_E\left(\frac{1}{2}(\delta s_E)^2 f^{''}(s_{E_0})\right)} - \beta_E V_E\left(\delta s_E f^{'}(s_{E_0})\right) \\ &\quad -\beta_E V_E\left(\frac{1}{2}(\delta s_E)^2 f^{''}(s_{E_0})\right) \\ \frac{dV_I}{dt} &= -\alpha_I V_I + (1-I_0)\beta_I f^{'}(s_{I_0})\delta s_I - \beta_I f(s_{I_0})V_I + \underbrace{(1-I_0)\beta_I\left(\frac{1}{2}(\delta s_I)^2 f^{''}(s_{I_0})\right)} - \beta_I V_I\left(\delta s_I f^{'}(s_{I_0})\right) \\ &\quad -\beta_I V_I\left(\frac{1}{2}(\delta s_I)^2 f^{''}(s_{I_0})\right). \end{aligned}$$

For a better convergence to the former Wilson-Cowan dynamics [23], the terms under the brace are discarded, and the remaining dynamics are

$$\begin{aligned} \frac{dV_E}{dt} &= -\alpha_E V_E + (1-E_0)\beta_E f^{'}(s_{E_0})\delta s_E - \beta_E f(s_{E_0})V_E - \beta_E V_E\left(\delta s_E f^{'}(s_{E_0})\right) - \beta_E V_E\left(\frac{1}{2}(\delta s_E)^2 f^{''}(s_{E_0})\right) \\ \frac{dV_I}{dt} &= -\alpha_I V_I + (1-I_0)\beta_I f^{'}(s_{I_0})\delta s_I - \beta_I f(s_{I_0})V_I - \beta_I V_I\left(\delta s_I f^{'}(s_{I_0})\right) - \beta_I V_I\left(\frac{1}{2}(\delta s_I)^2 f^{''}(s_{I_0})\right). \end{aligned}$$

Replacing $\delta s_E(t)$ and $\delta s_I(t)$ by their expressions we obtain

$$\begin{aligned} \frac{dV_E(t)}{dt} &= A_{EE}V_E(t) + A_{EI}V_I(t) + L_{EE}V_E^2(t) + L_{EI}V_E(t)V_I(t) + B_{1E}V_E^3(t) + B_{2E}V_E V_I^2(t) + B_{3E}V_I V_E^2(t) \\ &\quad + C_{E_0}A_E \sin(\omega_E t + \phi_E^0) + C_{E_1}V_E(t)A_E \sin(\omega_E t + \phi_E^0) + C_{E_{3a}}V_E(t)\big(A_E \sin(\omega_E t + \phi_E^0)\big)^2 \\ &\quad + C_{E_{3b}}V_E^2(t)A_E \sin(\omega_E t + \phi_E^0) + C_{E_{3c}}V_E(t)V_I(t)A_E \sin(\omega_E t + \phi_E^0), \\ \frac{dV_I(t)}{dt} &= A_{IE}V_E(t) + A_{II}V_I(t) + L_{II}V_I^2(t) + L_{IE}V_E(t)V_I(t) + B_{1I}V_I^3(t) + B_{2I}V_E V_I^2(t) + B_{3I}V_I V_E^2(t) \\ &\quad + C_{I_0}A_I \sin(\omega_I t + \phi_I^0) + C_{I_1}V_I(t)A_I \sin(\omega_I t + \phi_I^0) + C_{I_{3a}}V_I(t)\big(A_I \sin(\omega_I t + \phi_I^0)\big)^2 \\ &\quad + C_{I_{3b}}V_I^2(t)A_I \sin(\omega_I t + \phi_I^0) + C_{I_{3c}}V_E(t)V_I(t)A_I \sin(\omega_I t + \phi_I^0), \end{aligned}$$

Since we are interested in the effect of the low frequencies, we neglect the square terms $\left(A_E \sin(\omega_E t + \phi_E^0)\right)^2$ and $\left(A_E \sin(\omega_E t + \phi_E^0)\right)^2$ which involved higher harmonics at frequencies $2\omega_E$ and $2\omega_I$. By including Gaussian white noise, we finally obtain the following approximated stochastic dynamics

$$\begin{aligned}
\frac{dV_E(t)}{dt} &= A_{EE}V_E(t) + A_{EI}V_I(t) + L_{EE}V_E^2(t) + L_{EI}V_E(t)V_I(t) + B_{1E}V_E^3(t) + B_{2E}V_EV_I^2(t) + B_{3E}V_IV_E^2(t) \\
&\quad + \underbrace{C_{E_0}A_E \sin(\omega_E t + \phi_E^0)} + \underline{C_{E_1}V_E(t)A_E \sin(\omega_E t + \phi_E^0) + C_{E_{3b}}W_{EE}V_E^2(t)A_E \sin(\omega_E t + \phi_E^0)} \\
&\quad + \underline{C_{E_{3c}}W_{EI}V_E(t)V_I(t)A_E \sin(\omega_E t + \phi_E^0)} + \sigma_E \xi_E(t), \qquad (33) \\
\frac{dV_I(t)}{dt} &= A_{IE}V_E(t) + A_{II}V_I(t) + L_{II}V_I^2(t) + L_{IE}V_E(t)V_I(t) + B_{1I}V_I^3(t) + B_{2I}V_EV_I^2(t) + B_{3I}V_IV_E^2(t) \\
&\quad + \underbrace{C_{I_0}A_I \sin(\omega_I t + \phi_I^0)} + \underline{C_{I_1}V_I(t)A_I \sin(\omega_I t + \phi_I^0) + C_{I_{3b}}W_{II}V_I^2(t)A_I \sin(\omega_I t + \phi_I^0)} \\
&\quad + \underline{C_{I_{3c}}W_{IE}V_E(t)V_I(t)A_I \sin(\omega_I t + \phi_I^0)} + \sigma_I \xi_I(t), \qquad (34)
\end{aligned}$$

with the following parameter definitions:

$$\begin{aligned}
&A_{EE} = -\left[\alpha_E + \beta_E f(s_{E_0}) - (1-E_0)\beta_E f^{'}(s_{E_0})W_{EE}\right], \quad A_{EI} = -(1-E_0)\beta_E f^{'}(s_{E_0})W_{EI}, \\
&A_{II} = -\left[\alpha_I + \beta_I f(s_{I_0}) + (1-I_0)\beta_I f^{'}(s_{I_0})W_{II}\right], \quad A_{IE} = (1-I_0)\beta_I f^{'}(s_{I_0})W_{IE}, \\
&L_{EE} = -\beta_E f^{'}(s_{E_0})W_{EE}, \quad L_{EI} = \beta_E f^{'}(s_{E_0})W_{EI}, \quad B_{1E} = -\frac{1}{2}\beta_E f^{''}(s_{E_0})W_{EE}^2, \quad B_{2E} = -\frac{1}{2}\beta_E f^{''}(s_{E_0})W_{EI}^2, \\
&L_{IE} = -\beta_I f^{'}(s_{I_0})W_{IE}, \quad L_{II} = \beta_I f^{'}(s_{I_0})W_{II}, \quad B_{1I} = -\frac{1}{2}\beta_I f^{''}(s_{I_0})W_{II}^2, \quad B_{3E} = \beta_E f^{''}(s_{E_0})W_{EE}W_{EI}, \\
&B_{2I} = \beta_I f^{''}(s_{I_0})W_{IE}W_{II}, \quad B_{3I} = -\frac{1}{2}\beta_I f^{''}(s_{I_0})W_{IE}^2, \quad C_{E_0} = (1-E_0)\beta_E f^{'}(s_{E_0}), \quad C_{E_1} = -\beta_E f^{'}(s_{E_0}), \\
&C_{E_{3b}} = -\beta_E f^{''}(s_{E_0})W_{EE}, \quad C_{E_{3c}} = \beta_E f^{''}(s_{E_0})W_{EI}, \quad C_{I_0} = (1-I_0)\beta_I f^{'}(s_{I_0}), \quad C_{I_1} = -\beta_I f^{'}(s_{I_0}), \\
&C_{I_{3b}} = \beta_I f^{''}(s_{I_0})W_{II}, \quad C_{I_{3c}} = -\beta_I f^{''}(s_{I_0})W_{IE}.
\end{aligned}$$

The dynamics Eqs.9-10 constitute an approximate version of the Stochastic Wilson equations driven by slow periodic inputs h_E^{ext} and h_I^{ext}. We observe that the periodic inputs appear in these new dynamics in two ways. They are additive (terms underbrace) and multiplicative (terms underline). We are interested in the coupling between slow exogenous inputs and intrinsic rhythms. We first consider that the SWC model generates fast gamma-band rhythms when slow exogenous inputs have frequencies less than or equal to 10 Hz. We remark that the additive and multiplicative contributions of the exogenous inputs have different effects on the dynamics of the fast rhythms. The multiplicative contributions (terms underlined) act as parameters since the sensory inputs vary on a slow timescale compared to fast rhythms; they can be considered as constants during an oscillation period of the fast rhythms. Their primary effect is a slow modulation of the envelope, phase, and frequency of the fast rhythms. The additive contributions (terms under the underbrace) exert a forcing effect (depending on the amplitude of the sensory inputs) and drive the fast rhythms to evolve according to their own dynamics. The resulting dynamics may be interpreted as the superposition of these two effects. We treat them separately. First, the effects of the parametric terms are examined. Since their primary role is to modulate the fast-rhythm components, an appropriate framework is the envelope-phase decomposition. Such a framework will allow us to visualize how the envelope and the phase of the fast rhythms are directly coupled with the components of the slow exogenous inputs.

Envelope-Phase decomposition with only multiplicative or parametric contributions: Modulation

We look for the solutions of Eqs.9-10 without additive contributions (terms underbrace) in the following forms:

$$\begin{aligned}
V_E^{par}(t) &= Z\cos\left(\omega_0 t + \phi\right) \qquad (35) \\
V_I^{par}(t) &= \alpha Z\cos\left(\omega_0 t + \phi + \delta\right) \qquad (36)
\end{aligned}$$

where Z and ϕ are the slow envelope and phase variables of the fast rhythms and ω_0 its angular frequency defined as the imaginary part of the complex conjugate eigenvalues of deterministic dynamics of Eqs.9-10 (without exogenous inputs).

$$\omega_0 = \frac{1}{2}\sqrt{-(A_{EE} - A_{II})^2 - 4A_{EI}A_{IE}}\,. \tag{37}$$

The parameters α and δ defined as the amplitude ratio and phase difference between the E and I dynamics have the following expressions [23]:

$$\alpha = \sqrt{\frac{A_{IE}}{-A_{EI}}}; \quad \delta = \arctan\left(\frac{-2\omega_0}{A_{EE} - A_{II}}\right). \tag{38}$$

We replace the trial solutions Eqs.35-36 in Eqs.9-10 without additive contributions (terms underbrace) and look for the corresponding envelope-phase dynamics, which is obtained in the following forms.

$$\frac{dZ(t)}{dt} = F_1\big(Z,\phi\big) + G_1\big(Z,\phi,\xi_E,\xi_I\big) \tag{39}$$

$$\frac{d\phi(t)}{dt} = F_2\big(Z,\phi\big) + G_2\big(Z,\phi,\xi_E,\xi_I\big) \tag{40}$$

with the following functions:

$$\begin{aligned}
F_1\big(Z,\phi\big) &= \frac{1}{\alpha\sin(\delta)}\Big[\alpha f_1\big(Z,\phi\big)\sin(\omega_0 t+\phi+\delta) - f_2\big(Z,\phi\big)\sin(\omega_0 t+\phi)\Big] \\
F_2\big(Z,\phi\big) &= \frac{1}{\alpha Z\sin(\delta)}\Big[\alpha f_1\big(Z,\phi\big)\cos(\omega_0 t+\phi+\delta) - f_2\big(Z,\phi\big)\cos(\omega_0 t+\phi)\Big] \\
G_1\big(Z,\phi,\xi_E,\xi_I\big) &= \frac{1}{\alpha\sin(\delta)}\Big[\alpha g_1\big(Z,\phi,\xi_E,\xi_I\big)\sin(\omega_0 t+\phi+\delta) - g_2\big(Z,\phi,\xi_E,\xi_I\big)\sin(\omega_0 t+\phi)\Big] \\
G_2\big(Z,\phi,\xi_E,\xi_I\big) &= \frac{1}{\alpha Z\sin(\delta)}\Big[\alpha g_1\big(Z,\phi,\xi_E,\xi_I\big)\cos(\omega_0 t+\phi+\delta) - g_2\big(Z,\phi,\xi_E,\xi_I\big)\cos(\omega_0 t+\phi)\Big] \\
f_1\big(Z,\phi\big) &= \omega_0 Z\sin(\omega_0 t+\phi) + A_{EE}Z\cos(\omega_0 t+\phi) + A_{EI}\alpha Z\cos(\omega_0 t+\phi+\delta) + \\
&\quad L_{EE}Z^2\cos^2(\omega_0 t+\phi) + L_{EI}\alpha Z^2\cos(\omega_0 t+\phi)\cos(\omega_0 t+\phi+\delta) + B_{1E}Z^3\cos^3(\omega_0 t+\phi) + \\
&\quad B_{2E}\alpha^2 Z^3\cos(\omega_0 t+\phi)\cos^2(\omega_0 t+\phi+\delta) + B_{3E}\alpha Z^3\cos^2(\omega_0 t+\phi)\cos(\omega_0 t+\phi+\delta) \\
&\quad + C_{E_1}A_E\sin(\omega_E t+\phi_E^0)Z\cos(\omega_0 t+\phi) + C_{E_{3b}}A_E\sin(\omega_E t+\phi_E^0)Z^2\cos^2(\omega_0 t+\phi) \\
&\quad + C_{E_{3c}}A_E\sin(\omega_E t+\phi_E^0)\alpha Z^2\cos(\omega_0 t+\phi)\cos(\omega_0 t+\phi+\delta) \\
f_2\big(Z,\phi\big) &= \alpha\omega_0 Z\sin(\omega_0 t+\phi+\delta) + A_{IE}Z\cos(\omega_0 t+\phi) + A_{II}\alpha Z\cos(\omega_0 t+\phi+\delta) + L_{II}\alpha^2 Z^2 \times \\
&\quad \cos^2(\omega_0 t+\phi+\delta) + L_{IE}\alpha Z^2\cos(\omega_0 t+\phi)\cos(\omega_0 t+\phi+\delta) + B_{1I}\alpha^3 Z^3\cos^3(\omega_0 t+\phi+\delta) + \\
&\quad B_{2I}\alpha^2 Z^3\cos(\omega_0 t+\phi)\cos^2(\omega_0 t+\phi+\delta) + B_{3I}\alpha Z^3\cos^2(\omega_0 t+\phi)\cos(\omega_0 t+\phi+\delta) + \\
&\quad C_{I_1}A_I\sin(\omega_I t+\phi_I^0)\alpha Z\cos(\omega_0 t+\phi+\delta) + C_{I_{3b}}A_I\sin(\omega_I t+\phi_I^0)\alpha^2 Z^2\cos^2(\omega_0 t+\phi+\delta) \\
&\quad C_{I_{3c}}A_I\sin(\omega_I t+\phi_I^0)\alpha Z^2\cos(\omega_0 t+\phi)\cos(\omega_0 t+\phi+\delta) \\
g_1\big(Z,\phi,\xi_E,\xi_I\big) &= \sigma_E\xi_E(t), \quad g_2\big(Z,\phi,\xi_E,\xi_I\big) = \sigma_I\xi_I(t).
\end{aligned}$$

To obtain simple dynamics of the slow envelope and phase (Z,ϕ), we used the Stochastic Averaging Method (SAM) [1,25], which stipulates that the system can be approximated by a two-dimensional Markov process given by:

$$d\begin{pmatrix} Z(t) \\ \phi(t) \end{pmatrix} = \begin{pmatrix} m_1(t) \\ m_2(t) \end{pmatrix} dt + \begin{pmatrix} h_{11}(t) & h_{12}(t) \\ h_{21}(t) & h_{22}(t) \end{pmatrix}\begin{pmatrix} dW_1(t) \\ dW_2(t) \end{pmatrix}$$

with

$$
\begin{aligned}
m_i(X) &= T^{av}\Big(E\{F_i\} + \sum_j \int_{-\infty}^{0} E\Big\{\Big(\frac{\partial G_i}{\partial X_j}\Big)_t (G_j)_{t+\tau}\Big\} d\tau\Big) \\
\big[hh^{'}\big]_{ij} &= T^{av}\Big(\int_{-\infty}^{\infty} E\{(G_i)_t (G_j)_{t+\tau}\} d\tau\Big), \quad i,j = 1,2.
\end{aligned}
$$

Here $m = \begin{pmatrix} m_1(t) \\ m_2(t) \end{pmatrix}$, $F = \begin{pmatrix} F_1(t) \\ F_2(t) \end{pmatrix}$, $G = \begin{pmatrix} G_1(t) \\ G_2(t) \end{pmatrix}$, $X = \begin{pmatrix} Z_E(t) \\ \phi_E(t) \end{pmatrix}$,

h is the matrix made up of the elements h_{ij}, $(h\,h')$ is the product of h and its transpose. Moreover, $E.$ denotes the expectation operator (i.e. average over noise realizations), and T^{av} is the time averaging operator defined by

$$
T^{av}(.) = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} (.) dt\,.
$$

The subscripts t and $t+\tau$ mean that the elements of the corresponding vectors and matrix are evaluated at these times. The SAM further assumes that the 2-dimensional functions $F(Z,\phi)$ and $G(Z,\phi,\xi_E,\xi_I)$ are respectively of order $\mathrm{O}(\epsilon)$ and $\mathrm{O}(\epsilon^2)$, where ϵ is a small parameter. The averaging in the SAM procedure supposes that the functions $Z(t)$ and $\phi(t)$ are constant during the period $T_0 = 2\pi/\omega_0$ of the underlying deterministic oscillation (ω_0 also varies with system parameters). Furthermore, we also assume that the slow sensory inputs $A_E \sin(\omega_E t + \phi_E^0)$ and $A_E \sin(\omega_E t + \phi_E^0)$ are also constants during that same period. Therefore, these quantities are constant when we integrate over the period of the fast rhythms. After calculations, we end up with the following amplitude-phase dynamics:

$$
dZ(t) = \Big[-\Big(\nu + C_1^E A_E \sin(\omega_E t + \phi_E^0) + C_1^I A_I \sin(\omega_I t + \phi_I^0)\Big) Z(t) + B_1 Z^3(t) + \frac{D}{2Z(t)}\Big] dt + \sqrt{D} dW_1(t) \tag{41}
$$

$$
d\phi(t) = \Big[C_2^E A_E \sin(\omega_E t + \phi_E^0) + C_2^I A_I \sin(\omega_I t + \phi_I^0) + B_2 Z^2(t)\Big] dt + \frac{\sqrt{D}}{Z(t)} dW_2(t). \tag{42}
$$

The coefficients C_1^E, C_1^I, C_2^E and C_2^I depend on the parameters of the fast gamma network. Their expressions are given as follows

$$
C_1^E = \frac{1}{2}\beta_E f^{'}(s_{E_0}), \quad C_2^E = -\frac{\beta_E f^{'}(s_{E_0})}{2\alpha \sin(\delta)} \tag{43}
$$

$$
C_1^I = \frac{1}{2}\beta_I f^{'}(s_{I_0}), \quad C_2^I = -\frac{\beta_I f^{'}(s_{I_0})}{2\alpha \sin(\delta)} \tag{44}
$$

Other coefficients are given by [23]

$$
\nu = -\frac{A_{EE} + A_{II}}{2}, \quad D = \frac{(\sigma_I)^2 + (\alpha\sigma_E)^2}{2\alpha^2 \sin^2(\delta)}, \tag{45}
$$

$$
B_1 = \frac{1}{8}\Big[3B_{1E} + B_{3I} + \alpha^2(B_{2E} + 3B_{1I}) + 2\alpha\cos(\delta)(B_{3E} + B_{2I})\Big], \tag{46}
$$

$$
B_2 = \frac{1}{8\sin(\delta)}\Big[2\alpha(B_{3E} - B_{2I}) + 3(B_{1E} - B_{3I})\cos(\delta) + 3\alpha^2(B_{2E} - B_{1I})\cos(\delta) + \alpha(B_{3E} - B_{2I})\cos(2\delta)\Big]. \tag{47}
$$

However, these dynamics do not fully capture the SWC driven by sensory inputs. They describe only the parametric contributions of the exogenous inputs. To obtain a better approximation of the SWC driven by sensory inputs, we need to take into account the additive terms that have been discarded so far.

Role of the additive contributions

The additive terms act as forcing terms that tend to lock the fast dynamics to their own motion. This allows us to identify additive contributions in the following forms.

$$V_E^{add}(t) = a_1 \cos(\omega_E t) + b_1 \sin(\omega_E t) + c_1 \cos(\omega_I t) + d_1 \sin(\omega_I t) \tag{48}$$
$$V_I^{add}(t) = a_2 \cos(\omega_E t) + b_2 \sin(\omega_E t) + c_2 \cos(\omega_I t) + d_2 \sin(\omega_I t) \tag{49}$$

We focus on the weak amplitude of the exogenous inputs. We look for the values of the coefficients in Eqs. 48-49 by replacing them in the linear Wilson-Cowan equations driven by additive contributions only;

$$\frac{dV_E(t)}{dt} = A_{EE}V_E(t) + A_{EI}V_I(t) + C_{E_0}A_E \sin(\omega_E t + \phi_E^0) \tag{50}$$
$$\frac{dV_I(t)}{dt} = A_{IE}V_E(t) + A_{II}V_I(t) + C_{I_0}A_I \sin(\omega_I t + \phi_I^0) \tag{51}$$

After collecting coefficients of terms $\cos(\omega_E t)$, $\sin(\omega_E t)$, $\cos(\omega_I t)$ and $\sin(\omega_I t)$ and equating them to zero, we end up with the following eight dimensional linear system of equations.

$$\begin{bmatrix} \omega_E & A_{EE} & 0 & 0 & 0 & A_{EI} & 0 & 0 \\ A_{EE} & -\omega_E & 0 & 0 & A_{EI} & 0 & 0 & 0 \\ 0 & 0 & \omega_I & A_{EE} & 0 & 0 & 0 & A_{EI} \\ 0 & 0 & A_{EE} & -\omega_I & 0 & 0 & A_{EI} & 0 \\ 0 & A_{IE} & 0 & 0 & \omega_E & A_{II} & 0 & 0 \\ A_{IE} & 0 & 0 & 0 & A_{II} & -\omega_E & 0 & 0 \\ 0 & 0 & 0 & A_{IE} & 0 & 0 & \omega_I & A_{II} \\ 0 & 0 & A_{IE} & 0 & 0 & 0 & A_{II} & -\omega_I \end{bmatrix} \cdot \begin{bmatrix} a_1 \\ b_1 \\ c_1 \\ d_1 \\ a_2 \\ b_2 \\ c_2 \\ d_2 \end{bmatrix} = \begin{bmatrix} -C_{E_0}A_E \cos(\phi_E^0) \\ -C_{E_0}A_E \sin(\phi_E^0) \\ 0 \\ 0 \\ 0 \\ 0 \\ -C_{I_0}A_I \cos(\phi_I^0) \\ -C_{I_0}A_I \sin(\phi_I^0) \end{bmatrix} \tag{52}$$

The coefficients a_1, b_1,c_1,d_1, a_2, b_2,c_2 and d_2 can be obtained by solving the system of linear equations above. For the case in which the sensory inputs applied to the E and I populations have the same frequency ω, we may reduce the system described above as follows.

$$\begin{bmatrix} \omega & A_{EE} & 0 & A_{EI} \\ A_{EE} & -\omega & A_{EI} & 0 \\ 0 & A_{IE} & \omega & A_{II} \\ A_{IE} & 0 & A_{II} & -\omega_I \end{bmatrix} \cdot \begin{bmatrix} a_1 \\ b_1 \\ a_2 \\ b_2 \end{bmatrix} = \begin{bmatrix} -C_{E_0}A_E \cos(\phi_E^0) \\ -C_{E_0}A_E \sin(\phi_E^0) \\ -C_{I_0}A_I \cos(\phi_I^0) \\ -C_{I_0}A_I \sin(\phi_I^0) \end{bmatrix} \tag{53}$$

The systems can be solved using Cramer's method, which leads to

$$
\begin{aligned}
a_1 &= -\frac{C_{E_0}}{\Delta}\begin{vmatrix} -\omega & A_{EI} & 0 \\ A_{IE} & \omega & A_{II} \\ 0 & A_{II} & -\omega \end{vmatrix} A_E\cos(\phi_E^0) + \frac{C_{E_0}}{\Delta}\begin{vmatrix} A_{EE} & 0 & A_{EI} \\ A_{IE} & \omega & A_{II} \\ 0 & A_{II} & -\omega \end{vmatrix} A_E\sin(\phi_E^0) \qquad (54)\\
&\quad -\frac{C_{I_0}}{\Delta}\begin{vmatrix} A_{EE} & 0 & A_{EI} \\ -\omega & A_{EI} & 0 \\ 0 & A_{II} & -\omega \end{vmatrix} A_I\cos(\phi_I^0) + \frac{C_{I_0}}{\Delta}\begin{vmatrix} A_{EE} & 0 & A_{EI} \\ -\omega & A_{EI} & 0 \\ A_{IE} & \omega & A_{II} \end{vmatrix} A_I\sin(\phi_I^0)\,,\\
b_1 &= \frac{C_{E_0}}{\Delta}\begin{vmatrix} A_{EE} & A_{EI} & 0 \\ 0 & \omega & A_{II} \\ A_{IE} & A_{II} & -\omega \end{vmatrix} A_E\cos(\phi_E^0) - \frac{C_{E_0}}{\Delta}\begin{vmatrix} \omega & 0 & A_{EI} \\ 0 & \omega & A_{II} \\ A_{IE} & A_{II} & -\omega \end{vmatrix} A_E\sin(\phi_E^0) \qquad (55)\\
&\quad +\frac{C_{I_0}}{\Delta}\begin{vmatrix} \omega & 0 & A_{EI} \\ A_{EE} & A_{EI} & 0 \\ A_{IE} & A_{II} & -\omega \end{vmatrix} A_I\cos(\phi_I^0) - \frac{C_{I_0}}{\Delta}\begin{vmatrix} \omega & 0 & A_{EI} \\ A_{EE} & A_{EI} & 0 \\ 0 & \omega & A_{II} \end{vmatrix} A_I\sin(\phi_I^0)\,,\\
a_2 &= -\frac{C_{E_0}}{\Delta}\begin{vmatrix} A_{EE} & -\omega & 0 \\ 0 & A_{IE} & A_{II} \\ A_{IE} & 0 & -\omega \end{vmatrix} A_E\cos(\phi_E^0) + \frac{C_{E_0}}{\Delta}\begin{vmatrix} \omega & A_{EE} & A_{EI} \\ 0 & A_{IE} & A_{II} \\ A_{IE} & 0 & -\omega \end{vmatrix} A_E\sin(\phi_E^0) \qquad (56)\\
&\quad -\frac{C_{I_0}}{\Delta}\begin{vmatrix} \omega & A_{EE} & A_{EI} \\ A_{EE} & -\omega & 0 \\ A_{IE} & 0 & -\omega \end{vmatrix} A_I\cos(\phi_I^0) - \frac{C_{I_0}}{\Delta}\begin{vmatrix} \omega & A_{EE} & A_{EI} \\ A_{EE} & -\omega & 0 \\ 0 & A_{IE} & A_{II} \end{vmatrix} A_I\sin(\phi_I^0)\,,\\
b_2 &= \frac{C_{E_0}}{\Delta}\begin{vmatrix} A_{EE} & -\omega & A_{EI} \\ 0 & A_{IE} & \omega \\ A_{IE} & 0 & A_{II} \end{vmatrix} A_E\cos(\phi_E^0) - \frac{C_{E_0}}{\Delta}\begin{vmatrix} \omega & A_{EE} & 0 \\ 0 & A_{IE} & \omega \\ A_{IE} & 0 & A_{II} \end{vmatrix} A_E\sin(\phi_E^0) \qquad (57)\\
&\quad \frac{C_{I_0}}{\Delta}\begin{vmatrix} \omega & A_{EE} & 0 \\ A_{EE} & -\omega & A_{EI} \\ A_{IE} & 0 & A_{II} \end{vmatrix} A_I\cos(\phi_I^0) - \frac{C_{I_0}}{\Delta}\begin{vmatrix} \omega & A_{EE} & 0 \\ A_{EE} & -\omega & A_{EI} \\ 0 & A_{IE} & \omega \end{vmatrix} A_I\sin(\phi_I^0)\,.
\end{aligned}
$$

In the equations above, the quantity Δ is determinant defined as follows:

$$
\Delta = \begin{vmatrix} \omega & A_{EE} & 0 & A_{EI} \\ A_{EE} & -\omega & A_{EI} & 0 \\ 0 & A_{IE} & \omega & A_{II} \\ A_{IE} & 0 & A_{II} & -\omega_I \end{vmatrix}\,.
$$

We recall that the notation $\left|a_{i,j}\right|$ signifies the determinant of the matrix $(a_{i,j})$. If the theory assumes linear Wilson-Cowan dynamics, it is accurate in the quasi-cycle regime. Nevertheless, it also performs well in the limit-cycle regime near the Hopf bifurcation.

A complete solution of the approximate SWC driven by slow exogenous inputs is obtained by adding the parametric and additive contributions as follows

$$
\begin{aligned}
V_E(t) &= V_E^{par}(t) + V_E^{add}(t) = Z\cos\left(\omega_0 t + \phi\right) + a_1\cos(\omega_E t) + b_1\sin(\omega_E t) + c_1\cos(\omega_I t) + d_1\sin(\omega_I t) \qquad (58)\\
V_I(t) &= V_I^{par}(t) + V_I^{add}(t) = \alpha Z\cos\left(\omega_0 t + \phi + \delta\right) + a_2\cos(\omega_E t) + b_2\sin(\omega_E t) + c_2\cos(\omega_I t) + d_2\sin(\omega_I t) \qquad (59)
\end{aligned}
$$

where the coefficients a_1, b_1,c_1,d_1, a_2, b_2,c_2 and d_2 are solutions of the system of equations 52. For the case where the network is driven by sensory inputs at the same frequency, the solution above is reduced to

$$
\begin{aligned}
V_E(t) &= V_E^{par}(t) + V_E^{add}(t) = Z\cos\left(\omega_0 t + \phi\right) + a_1\cos(\omega t) + b_1\sin(\omega t) \qquad (60)\\
V_I(t) &= V_I^{par}(t) + V_I^{add}(t) = \alpha Z\cos\left(\omega_0 t + \phi + \delta\right) + a_2\cos(\omega t) + b_2\sin(\omega t) \qquad (61)
\end{aligned}
$$

this time the coefficients are obtained from the expressions Eqs.54-57.

The Modulation Index

To compute the modulation index (MI), we first defined the amplitude-phase distribution as follows

- We consider the amplitude envelope of the fast component of the signal. This can be achieved using the Hilbert transform when the signal is described by the SWC (after filtering in the gamma band) or by the parametric dynamics of LFPs. For the SAM, the envelope-amplitude dynamics have been derived.
- the phase of the slow signal can be obtained easily since its form is sinusoidal.
- The phases ϕ_s of the slow signals are binned, and the mean of the amplitude-envelope of the fast signal over each bin is computed.
- The amplitude-phase distribution can be calculated as

$$P(j) = \frac{\left\langle Env\left(V_{E_f}^{par}\right)\right\rangle_{\phi_s}(j)}{\sum_{k=1}^{N}\left\langle Env\left(V_{E_f}^{par}\right)\right\rangle_{\phi_s}(k)} \tag{62}$$

The Modulation Index (MI) is defined as the Kullback-Leibler distance between the amplitude phase P and the uniform distributions. This leads to

$$MI = \frac{\log(N) - H(P)}{\log(N)} \tag{63}$$

where,

$$H(P) = -\sum_{j=1}^{N} P(j)\log\Big(P(j)\Big)$$

Spectral Analysis

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